**6.** (i) The curve $C_1$ has equation

$$y = 3\ln(x^2 - 5) - 4x^2 + 15 \qquad x > \sqrt{5}$$

Show that $C_1$ has a stationary point at $x = \dfrac{\sqrt{p}}{2}$ where $p$ is a constant to be found.

**(4)**

(ii) A different curve $C_2$ has equation

$$y = 4x - 12\sin^2 x$$

(a) Show that, for this curve,

$$\frac{\mathrm{d}y}{\mathrm{d}x} = A + B\sin 2x$$

where $A$ and $B$ are constants to be found.

(b) Hence, state the maximum gradient of this curve.

**(4)**